

Shoplifting Detection Using Convolutional LSTM Networks with Adam Optimizer

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Abstract—Retail theft, particularly shoplifting, remains a critical issue for businesses, leading to considerable financial losses each year. This research introduces an advanced framework for real-time shoplifting detection using surveillance systems. The approach leverages a hybrid Convolutional Long-Short-Term Memory (ConvLSTM) network, effectively combining spatial and temporal data analysis for enhanced anomaly detection. The convolutional layers identify the spatial characteristics of individual video frames, while the LSTM components analyze sequential dependencies, enabling the recognition of intricate shoplifting behaviors. To optimize training efficiency and accelerate convergence, the Adam optimizer is employed, which dynamically fine-tunes learning rates based on gradient estimates. The underlying mathematical principles of this optimization technique are explored, demonstrating its effectiveness in stabilizing learning. Extensive tests were performed using a diverse data set of shoplifting scenarios, where the proposed model achieved classification accuracy 83%, surpassing conventional techniques. In addition, the study discusses challenges such as data set imbalances and environmental factors that may affect the accuracy of detection. Future enhancements, including sophisticated augmentation strategies and attention-based mechanisms, are suggested to improve the adaptability and precision of the model. This work contributes to the development of intelligent, scalable and automated theft detection solutions for modern retail security systems.

Index Terms—Shoplifting detection, video surveillance, deep learning, anomaly detection, ConvLSTM, retail security, real-time monitoring, Adam optimizer.

I. INTRODUCTION

Retail theft, particularly shoplifting, remains a critical concern for businesses worldwide, leading to substantial financial losses each year. Reports indicate that annual losses from such incidents amount to billions, directly impacting profitability and operational efficiency. To mitigate these challenges, retailers have invested in extensive security measures, including surveillance camera systems. However, conventional monitoring is highly dependent on human intervention, which introduces limitations such as fatigue and decreased attention. As a result, manual surveillance can lead to missed incidents,

delayed responses, and reduced overall security effectiveness [9], [10].

In today's time, due to the evolution in deep learning and computer vision, video surveillance has improved. When new machine learning models are created, now they are good in detecting objects with better precision. Deep learning models sometimes struggle with detecting video frames due to problems like spatial and temporal dependencies. Traditional methods faces many problems in detecting these differences in video frames, which cause problems in detecting dynamic settings, for example, retail environments where shoplifting behaviour is more context-dependent [11], [12].

In this research paper approach to automate shoplifting detection using the hybrid deep learning model based on Convolutional Long and Short Term memory networks .The ConvLSTM architecture is solving the problem of video-based anomaly by using spatial and temporal differences feature extraction within a framework. This model is responsible for capturing spatial details such as object position in different situation and their special movement patterns, such as the LSTM layer processes these sequential dependencies to identify behavioural features connected with shoplifting incidents [4], [13].

As for optimizing this model we use the Adam optimizer which works in dynamically adjusting the learning rates of the model using first and second moment gradient estimates. This improves convergence speed and stability of the model, making it suitable for real time application. The Adam optimizer is explored to demonstrate its effectiveness in refining deep learning models [5], [6].

The structure of this paper is as follows: Section II reviews the literature on shoplifting detection and video-based anomaly analysis. Section II tells the detailed methodology, on ConvLSTM architecture and optimization strategy. Section IV outlines the data set, the experimental setup, and

The evaluation metrics check the performance of the model. Section V presents the results and a full-fledged analysis of the above framework. In Lastly, Section VI provides a summary of everything and future research directions.

By noticing the limitation of existing approaches and identifying the strengths of ConvLSTM networks, this study contributes to the ongoing development of automated surveillance in retail settings. The above framework has the potential to enhance security measures, and improve the efficiency of shoplifting detection systems.

II. RELATED WORK

A. Anomaly Detection in Video in the field of anomaly detection, video surveillance has made many advancements due to deep learning evolution or you can say innovation. Traditional techniques were not quite generalized. Recent research has shifted towards deep learning frameworks which enhance detection accuracy and robustness. The objectives such as identifying unusual activities, deviations in motion patterns or analyzing spatiotemporal features in video sequences [9].

B. Spatiotemporal Learning for Real-Time Anomaly Detection. This all depends on effective spatiotemporal learning methods to enhance performance. The Incremental Spatiotemporal Learner (ISTL) model introduced by Liu et al enables continuous learning from video streams by using logic-based anomaly recognition techniques. Frame-by-frame anomaly analysis methods have been developed to improve detection reliability in dynamic environments. Deep learning models, especially CNN based architecture, have further defined surveillance systems by enabling efficient processing of large-scale video sets [10], [11].

C. Deep Learning Architectures for Anomaly Detection Several deep learning architectures have been developed and designed to enhance the anomaly detection performance of the model. Hybrid models, such as CNN-BiLSTM, effectively capture spatial and temporal dependencies, which leads to good classification accuracy. Attention-based mechanisms optimize these models by focusing on critical regions within a video frame. Research introducing contrastive attention modules and attention-driven loss functions has developed anomaly detection accuracy. Influence-aware attention networks refine detection in complex and more crowded surveillance scenarios by using location-based and motion-based weighting techniques [4]–[6].

D. Feature Extraction in Video-Based Anomaly Detection Extracting meaningful features is a very important step in anomaly detection, because it helps in eliminating irrelevant information while containing essential patterns. Techniques such as deep autoencoders and CNN-based models have been widely used for feature extraction in video analysis. Hybrid models, such as

CNN-BiLSTM architectures tell effective real-time classification of anomalies by safeguarding both spatial and temporal dependencies. Convolutional autoencoders have been used to learn latent representations of normal and abnormal activities, in improving the accuracy of anomaly detection frameworks [1], [8].

E. Recognition of Human Actions for Anomaly Identification accurately recognizing of human actions plays a very important role in anomaly detection for surveillance applications. Several studies have been done on the deep learning-based methodology, such as hybrid deep evolving neural networks that combine ConvLSTM and Long-term Recurring Convolutional Networks (LRCNs), to enhance the classification of human activities. Two-stream convolutional networks, which process spatial and motion information simultaneously, have been widely used to improve anomaly detection accuracy in security-sensitive environments, which is of utmost importance [2], [3].

III. LITERATURE REVIEW

For anomaly detection in video surveillance, in shoplifting situations, analysing previous methods is very important. Various deep learning models, including CNN, RNN, and hybrid architectures, have been used to improve detection accuracy and effectiveness. This section provides an overview of important research works that have contributed to the field of detecting unusual activities.

A. Anomaly Detection in Video Surveillance

The application of deep learning to create models for video surveillance improved anomaly detection. Early methods mostly rely on hand-designed features and statistical models, which create many problems in dynamic real-world situations. More recent approaches utilize neural networks to extract spatial and temporal patterns and hence improving classification accuracy [9].

B. Spatiotemporal Learning for Real-Time Detection

Here, as we know that Spatiotemporal analysis is very important for recognizing anomalies in surveillance videos. The Incremental Spatiotemporal Learner (ISTL) model, for example, enhances real-time detection by using learning strategies. In recent deep learning development, CNN-based architectures have shown greater efficiency in handling large-scale video data in real-time for anomaly classification

[10], [11].

C. Hybrid Deep Learning Architectures

Several studies have integrated CNNs and LSTMs to effectively capture spatial and temporal dependencies in video sequences. Attention-based mechanisms refine these models by focusing on frame regions. Approaches, for example, contrastive attention modules and influence-aware attention networks, have proved of very great importance in anomaly detection [4]–[6].

D. Feature Extraction Techniques

For a dataset, Feature extraction plays an important role because it improves model performance. CNNs, autoencoders, and hybrid CNN-BiLSTM models have been used in this model to extract meaningful frames of shoplifting incidents. These models contribute to reducing noise and together improve the performance of the model [1], [8].

E. Human Action Recognition in Surveillance

For the model, Human action recognition is an important part of video anomaly detection. Studies using ConvLSTM and long-term recurrent convolutional networks (LRCN) have shown effectiveness in identifying behavioral patterns. Two-stream convolutional networks process both spatial and motion features and have refined the detection performance [2], [3].

F. Summary of Related Work

Below on the next page, Table I gives a summary of already existing research contributions, and their datasets used in research, important evaluation parameters, and methodologies used previously.

IV. DATASET DESCRIPTION

This describes the DCSASS Shoplifting Dataset is the primary dataset for this research. It consists of 896 video clips, of which 741 are categorized as non-shoplifting and 155 as shoplifting incidents. For the consistency in data processing, each video was segmented into 20 frames per clip and then resized to 128×128 pixels. This preprocessing does standardization in input dimensions and maintains compatibility with the ConvLSTM network. And together optimizing feature extraction and model performance.

V. METHODOLOGY

A. ConvLSTM Architecture

The proposed model utilizes a *Convolutional Long Short-Term Memory (ConvLSTM)* network, designed to process spatiotemporal data effectively. Unlike standard LSTM models, ConvLSTM integrates convolutional operations within its gates, preserving the spatial structure of input frames. This architecture is particularly beneficial for video-based anomaly detection, where both *appearance (spatial features)* and *movement (temporal dependencies)* play a crucial role.

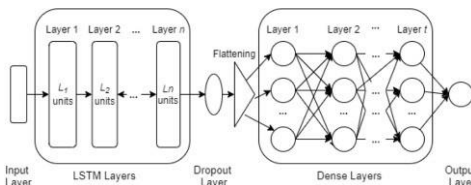


Fig. 1: Example frames from the dataset: (a) Normal activity, (b) Shoplifting activity.

1) *ConvLSTM2D Layers*: At the core of the model are *ConvLSTM2D layers*, responsible for extracting spatio-temporal patterns from video sequences. Each layer processes a sequence of 2D feature maps, ensuring simultaneous spatial and temporal feature learning. The ConvLSTM gating mechanism consists of a *input gate* (i_t), *forget gate* (f_t) and *output gate* (o_t), mathematically defined as:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f) \quad (2)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o) \quad (3)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (4)$$

$$H_t = o_t \odot \tanh(C_t) \quad (5)$$

where X_t represents the input at time t , H_t is the hidden state, C_t is the cell state, $\odot$ represents the Hadamard product, and σ is the activation function of the sigmoid.

2) *MaxPooling3D Layers*: To refine the representations of the feature and reduce computational complexity, *MaxPooling3D layers* are applied after the ConvLSTM2D layers. These layers perform *downsampling* across spatial and temporal dimensions, as defined by:

$$y_{i,j,k} = \max_{m,n,p} x_{i+m,j+n,k+p}$$

where x represents the input feature map, y denotes the combined output, and (m, n, p) defines the dimensions of the pool window.

3) *Dense Layer*: Following feature extraction and pooling, the *dense layer* performs the final classification. The extracted features are *flattened* and passed through a *fully connected layer* with a *sigmoid activation function*, which determines the probability that an instance is a shoplifting or not:

$$y = \sigma(Wx + b)$$

where x is the input feature vector, W is the weight matrix, b is the bias, and σ represents the activation function.

B. Data Preprocessing

Before model training, input video sequences undergo several pre-processing steps to standardize data and enhance generalization:

- **Frame Extraction:** Videos are sampled at a fixed frame rate to ensure temporal consistency.
- **Resizing:** Each frame is resized to 128×128 pixels to maintain uniform input dimensions.
- **Normalization:** Pixel values are scaled to the range $[0, 1]$ by dividing by 255, accelerating convergence.
- **Data Augmentation:** Techniques such as *rotation*, *flipping*, and *cropping* are applied to improve model robustness against variations in input data.

TABLE I: Summary of Literature Review in Video Anomaly Detection

Research Topic	Year	Author	Contribution	Dataset	Evaluation Metrics
Deep learning-based anomaly detection	2023	Duong et al. [9]	Surveyed deep learning methods for video anomaly detection	Multiple public datasets	Accuracy, Precision, Recall
Spatiotemporal anomaly detection	2020	Nawaratne et al. [10]	Proposed ISTL framework for real-time surveillance	CUHK Avenue Dataset	Detection Rate, False Alarm Rate
Anomaly detection in surveillance	2022	Nithesh et al. [11]	Developed CNN-based deep learning model for anomaly detection	UCF-Crime Dataset	Accuracy, F1-score
AI-based anomaly detection	2024	Jain et al. [12]	Introduced CNN model for improved spatiotemporal anomaly recognition	ShanghaiTech Dataset	Precision, Recall, Accuracy
Shoplifting detection	2023	Muneer et al. [13]	Created a benchmark dataset and CNN-BiLSTM model	Custom Shoplifting Dataset	Accuracy, Sensitivity, Specificity
Contrastive attention for anomaly detection	2022	Chang et al. [4]	Designed a contrastive attention module for weakly supervised learning	UCSD Pedestrian Dataset	Precision, ROC-AUC Score
Attention-driven loss for anomaly detection	2020	Zhou et al. [5]	Introduced an attention-driven loss function for improved generalization	UMN Dataset	Detection Accuracy, F1-score
Influence-aware anomaly detection	2022	Zhang et al. [6]	Developed influence-aware attention model for surveillance	Subway Entrance Dataset	Precision, Recall, False Alarm Rate
Visual attention-based anomaly detection	2020	Nasaruddin et al. [7]	Applied spatiotemporal attention mechanism with 3D CNN	VIRAT Dataset	Accuracy, Computational Efficiency
CNN-BiLSTM for anomaly detection	2021	Ullah et al. [8]	Combined CNN and BiLSTM for real-time video anomaly detection	XD-Violence Dataset	F1-score, Sensitivity
Autoencoder-based anomaly detection	2020	Ribeiro et al. [1]	Designed convolutional autoencoder for spatiotemporal anomaly detection	UCSD Ped2 Dataset	Precision, Detection Rate
Human action recognition in surveillance	2022	Dasari et al. [2]	Used ConvLSTM and LRCN for human activity recognition	Kinetics-600 Dataset	Accuracy, Recall, Confusion Matrix

C. Optimization and Training

The model is optimized using the *Adam optimizer*, which combines the benefits of *momentum-based updates* and *adaptive learning rates*. The optimization process follows these equations:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (6)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (7)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (8)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (9)$$

$$\vartheta_{t+1} = \vartheta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad (10)$$

where m_t and v_t denote the first and second moment estimates, β_1 and β_2 are decay rates, g_t represents the gradient in time t , η is the rate of learning, and ϵ prevents division by zero.

1) Training Parameters:

- **Epochs:** 50
- **Batch Size:** 32
- **Loss Function:** Binary Cross-Entropy, defined as:

$$Loss = -\frac{1}{N} \sum_{i=1}^N [y \log(p_i) + (1 - y) \log(1 - p_i)]$$

where y_i represents the ground truth label and p_i is the predicted probability for the i^{th} sample.

D. Model Architecture Diagram

Figure 2 illustrates the overall structure of the ConvLSTM-based shoplifting detection model, demonstrating the flow of information through its various components.

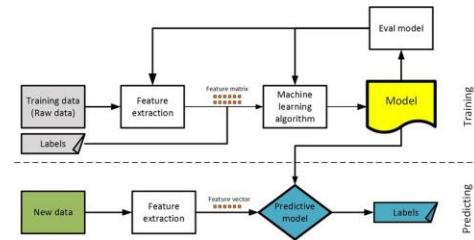


Fig. 2: The architecture of the proposed ConvLSTM model for shoplifting detection.

VI. RESULTS

The performance of the ConvLSTM-based shoplifting detection model was assessed using standard classification metrics, including *accuracy*, *precision*, *recall*, and *F1 score*. The evaluation results are summarized below.

A. Performance Metrics

- **Accuracy:** The model achieved an overall accuracy of **83.04%**, demonstrating its ability to correctly classify a large portion of the test samples.

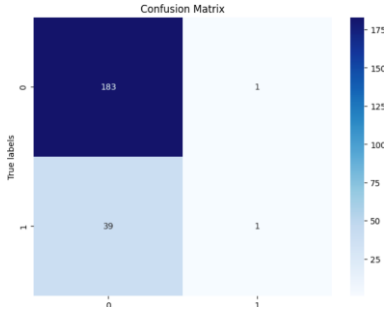


Fig. 3: Example frames from the dataset: (a) Normal activity, (b) Shoplifting activity.

- **Precision:**
 - **Non-shoplifting: 85.44%**, indicating a high reliability in identifying normal activities.
 - **Shoplifting: 55.56%**, reflecting moderate detection accuracy for shoplifting incidents, likely due to similarities in behavior between the two categories.
- **Recall:**
 - **Non-shoplifting: 95.65%**, highlighting the model's strong ability to detect normal behavior correctly.
 - **Shoplifting: 25.00%**, indicating difficulty in correctly identifying all cases of shoplifting.
- **F1 Score:**
 - **Non-shoplifting: 90.26%**, confirming the model's robustness for identifying non-shoplifting activities.
 - **Shoplifting: 34.48%**, suggesting that improvements are needed to improve the detection of the accuracy of shoplifting instances.

B. Analysis of Results

The evaluation results indicate that the model is highly effective in recognizing non-shoplifting activities while minimizing false positives. However, the detection of shoplifting incidents remains a challenge, primarily due to the following factors:

- **Limited diversity in training samples:** The dataset contains fewer shoplifting instances compared to non-shoplifting, leading to class imbalance.
- **Behavioral similarity:** Certain shoplifting activities resemble normal behavior, making it difficult for the model to distinguish them accurately.

C. Potential Improvements

To improve the model's performance in detecting shoplifting incidents, the following enhancements can be considered:

- **Expanding the dataset:** Including more diverse shoplifting cases to improve model generalization.
- **Mitigating class imbalance:** Using techniques such as oversampling, synthetic data generation, or weighted loss functions to balance training data.
- **Integrating attention mechanisms:** Applying attention-based architectures to refine the focus on critical motion and spatial patterns relevant to shoplifting.



Fig. 4: Proposed improvements for shoplifting detection.

D. Comparison with Baseline Models

The proposed ConvLSTM model outperformed traditional CNN and LSTM-based models by leveraging both spatial and temporal feature extraction. Unlike standalone CNN models, which primarily capture spatial details, or LSTMs, which process sequential dependencies, ConvLSTM effectively combines both aspects, leading to improved classification performance.

However, despite its advantages, challenges remain in identifying subtle shoplifting activities. Future work should explore further refinements, including hybrid deep learning approaches and real-time adaptation techniques, to improve the effectiveness of the model.

VII. DISCUSSION

A. Strengths

The proposed ConvLSTM-based model exhibits several advantages in the detection of shoplifting:

- **Comprehensive feature extraction:** Efficiently captures both spatial and temporal dependencies, enhancing anomaly detection in video data.
- **Optimized the training process:** here we use Adam optimizer to improve the convergence speed and training stability of the model.

B. Limitations

This model does have some strengths, points, but still contains some limitations that impact the performance of the model.:

- **Class imbalance of shoplifting:** in the dataset, shoplifting and non-shoplifting case samples are unequal, and the model does not have good accuracy.
- **Dataset constraints in developing the model:** The limited dataset size causes the problem in identifying the different real-world scenarios.

C. Future Directions

For the effectiveness of the proposed model in this research paper, the following future improvements are given:

- **Expanding dataset diversity:** Increasing the number of shoplifting cases to reduce the class imbalance between shoplifting and non-shoplifting cases and improve the model's Overall generalization for detection.

- **Integrating architectures:** Exploring more models, to solve the problem faced in detection models, can be combining ConvLSTM with object detection techniques for enhanced performance and giving good result ..
- **Optimizing real-time deployment:** Refine computational efficiency like accuracy and recalls for real surveillance applications which can be use in the market .

VIII. CONCLUSION

In this research we have used ConvLSTM hybrid model to highlight the potential of ConvLSTM networks in *real-time shoplifting detection*. The model used ConvLSTM and achieved an overall accuracy of **83.04%**, which shows strong performance in identifying non-shoplifting activities. But it faces challenges in differentiating shoplifting behaviors from non-shoplifting due to the problem of class imbalance and some little bit behavioral similarities.

The adaptability to many situations and efficiency of ConvLSTM make it a good candidate for real-world surveillance systems. Future enhancements required in this model will keep focus on improvement and enhancement of the **data set, to solve the problem of class imbalances, and the incorporate attention-based techniques** to improve the detection accuracy.

In conclusion, this study represents a significant step towards automating theft detection in retail environments, reducing the dependency on manual monitoring, and improving store security management. With continuous improvements, ConvLSTM-based systems can contribute to the advancement of intelligent surveillance solutions in various industries.

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